

Introduction to Windows Lightweight Virtual Machine (FVM)

1. Overview

FVM is a lightweight virtualization technology built on resource isolation mechanisms within the operating system kernel. Unlike traditional hardware virtualization architectures, FVM does not need to emulate the complete hardware stack. It can create multiple mutually isolated execution environments on a single physical host, achieving a balance between efficient resource utilization and secure isolation.

2. Functions and Terminology

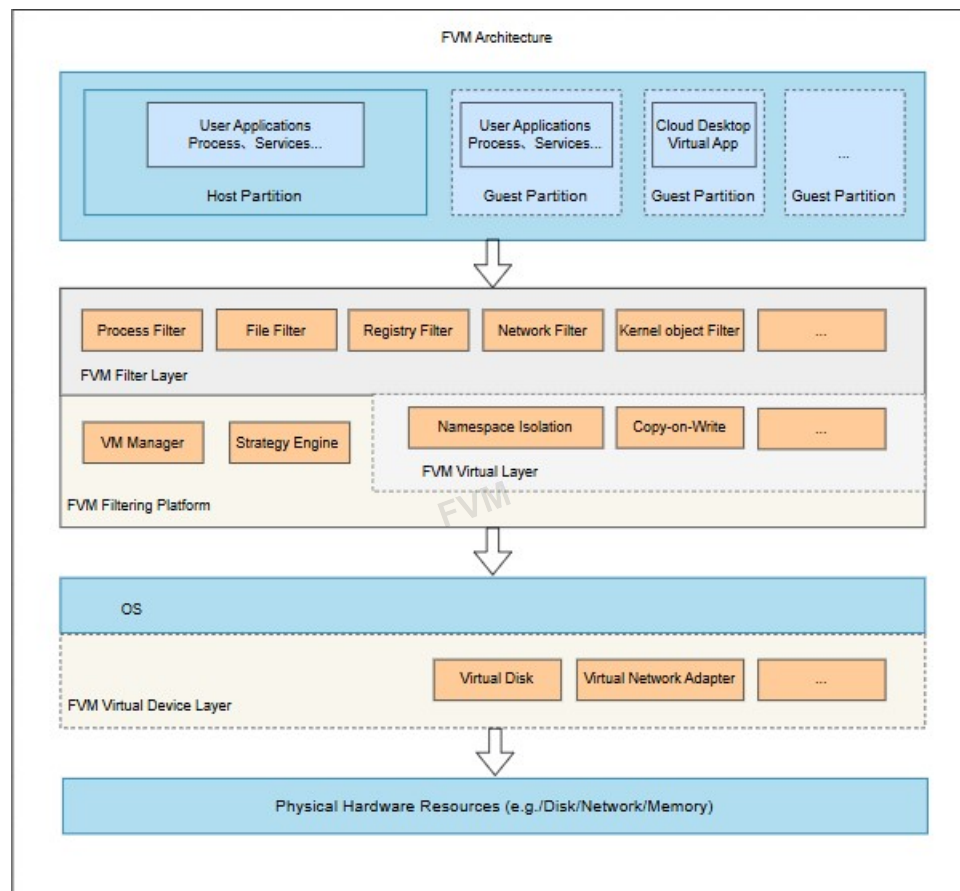
- **Host Partition :** The real physical operating system environment with full system privileges.
- **Guest Partition :** An application or process group running within a restricted, isolated virtual environment. It is not a complete operating system.
- **Namespace Isolation:** Guest partitions use private namespaces to achieve isolation between themselves and the host partition, as well as between different guest partitions.
- **Copy-on-Write :** A resource management technique where processes in the host and guest partitions share a read-only copy of the same resource. Only when a process in a guest partition needs to modify that resource is a private copy created for it.
- **Virtual Disk:** A software-emulated disk device, stored as a file on physical storage media, providing a complete block device interface to partitions.
- **Virtual Network Adapter:** A software-emulated network interface card, presented as a standard network device within a partition but actually communicating with the physical network through the virtualization layer.
- **Cloud Desktop Virtual Application:** Virtualizing multiple shared desktop partition environments on a single host machine, allowing multiple FVM guest partitions to be remotely connected and operated simultaneously.

3. Architectural Design

FVM constructs a virtualization layer between the operating system kernel and the application layer, employing namespace isolation and Copy-on-Write mechanisms. This design allows partition processes to directly share host system resources while ensuring complete isolation of the runtime environment. Each guest partition only needs to record its differences from the host environment, eliminating the need to duplicate the entire operating system image, significantly reducing storage and operational overhead.

4. System Architecture

The diagram below provides a brief overview of the FVM environment's architecture.



Host Partition and Guest Partition

FVM achieves isolation through partitions. A partition is a logical isolation unit provided by the virtual machine manager. Each partition runs a group of processes and is uniformly managed by the FVM Filtering Platform. Processes can join a specified partition or migrate between different partitions. Partitions possess capabilities such as resource limitation, namespace isolation, policy arbitration, monitoring, and auditing. An FVM virtualization program must have at least one partition running in the host. The virtualization management stack runs in the host partition and has direct access to system resources.

The host partition, by default, is not processed by the virtualization layer and can directly access system resources. Conversely, guest partitions are under the control of the FVM virtual layer by default. Guest partitions can read resources from the host partition but cannot write directly; all write requests are redirected by the FVM virtual layer to virtual devices for processing. Simultaneously, based on FVM's namespace isolation mechanism, the host partition cannot access resources within guest partitions, and guest partitions are also isolated from each other.

FVM Filtering Platform

The FVM Filtering Platform is the processing platform that controls how applications in each partition access system resources. It consists of the Filter Layer, Virtual Machine Manager, Strategy Engine, and Virtual Layer.

Filter Layer

The Filter Layer is a set of policy hooks attached to the system resource stack. It is responsible for monitoring, intercepting, and modifying resource access requests before they reach their original targets. This allows for extending or replacing the functions of original targets and provides instruction transmission and execution capabilities for the Virtual Layer and Strategy Engine.

The Filter Layer consists of the following core components:

- **Process Filter:** Monitors, intercepts, and allows process creation and termination. It determines which virtual machine partition a process belongs to, enabling process-level isolation and control.
- **File Filter:** Monitors, intercepts, and allows file access operations. It redirects file access paths, providing a virtualized file view for partitions.
- **Registry Filter:** Monitors, intercepts, and allows registry access operations. It redirects registry access paths, providing a virtualized registry view for partitions.
- **Network Filter:** Monitors, intercepts, and allows behaviors such as network connections and data transmission/reception. It supports control actions like traffic mirroring, access blocking, and transparent proxying.
- **Kernel Object Filter:** Monitors, intercepts, and allows access to kernel resources such as inter-process communication objects, synchronization objects, and handle operations. It redirects kernel resource access paths, providing a virtualized kernel resource view for partitions.
- **MISC Filter:** Monitors, intercepts, and modifies access behaviors to system resources like services, desktops, and memory, completing the final piece of resource control.

Virtual Machine Manager

The Virtual Machine Manager is the core management center of the FVM Filtering Platform. Its main responsibilities include:

- **Instance Lifecycle Management:** Responsible for creating, destroying, and maintaining the running state of partition instances, ensuring the entire process from startup to termination is controllable and traceable.

- **Resource Allocation and Limitation:** Allocates computing, storage, and network resources to partitions and enforces corresponding resource quota limits.
- **Policy Management and Enforcement:** Deploys and enforces various isolation policies, security policies, and access control rules.

Strategy Engine

The Strategy Engine is the core decision-making hub of the FVM Filtering Platform. It is responsible for the real-time determination of which partition a process belongs to and for fine-grained control over the resource access behavior of partition processes. Its core responsibilities include:

- **Execution Policy Judgment:** Evaluates and makes decisions on process behavior based on preset rules and real-time context.
- **Output Decision Results:** Generates clear instructions such as allow, deny, or redirect to guide the underlying execution units.
- **Security Logic Abstraction:** Extracts access control, permission verification, and other security logic from business code, centrally managing it within the Strategy Engine.
- **Rapid Policy Iteration:** Enables policy updates to keep pace with the evolving risk landscape through mechanisms like rule hot-loading and canary releases.

By completely decoupling policy from business, the FVM Filtering Platform achieves agile and deterministic security responses in complex, dynamic environments.

Virtual Layer

The Virtual Layer is the resource proxy layer of the FVM Filtering Platform. Relying on mechanisms like namespace isolation and copy-on-write, it achieves mutual isolation between partitions. Its core responsibilities include:

- **Resource Abstraction and Provisioning:** Provides virtualized views of computing, storage, and network resources to partition instances, shielding underlying physical topology differences and enabling on-demand resource allocation and dynamic adjustment.
- **Isolation Boundary Construction:** Isolates process views based on namespaces, maintains the integrity of partition boundaries, and prevents privilege escalation and illegal cross-boundary access.
- **Efficient Storage and Access Redirection:** Builds the partition's root file system and registry view using copy-on-write technology. Redirects access paths for resources like files and registry to the partition's dedicated storage area, achieving rapid deployment and efficient storage reuse.

- **Data Path Transformation :** Supports advanced data plane capabilities such as transparent proxying, traffic mirroring, and data encryption, providing underlying support for security control and policy execution.

FVM Virtual Device Layer

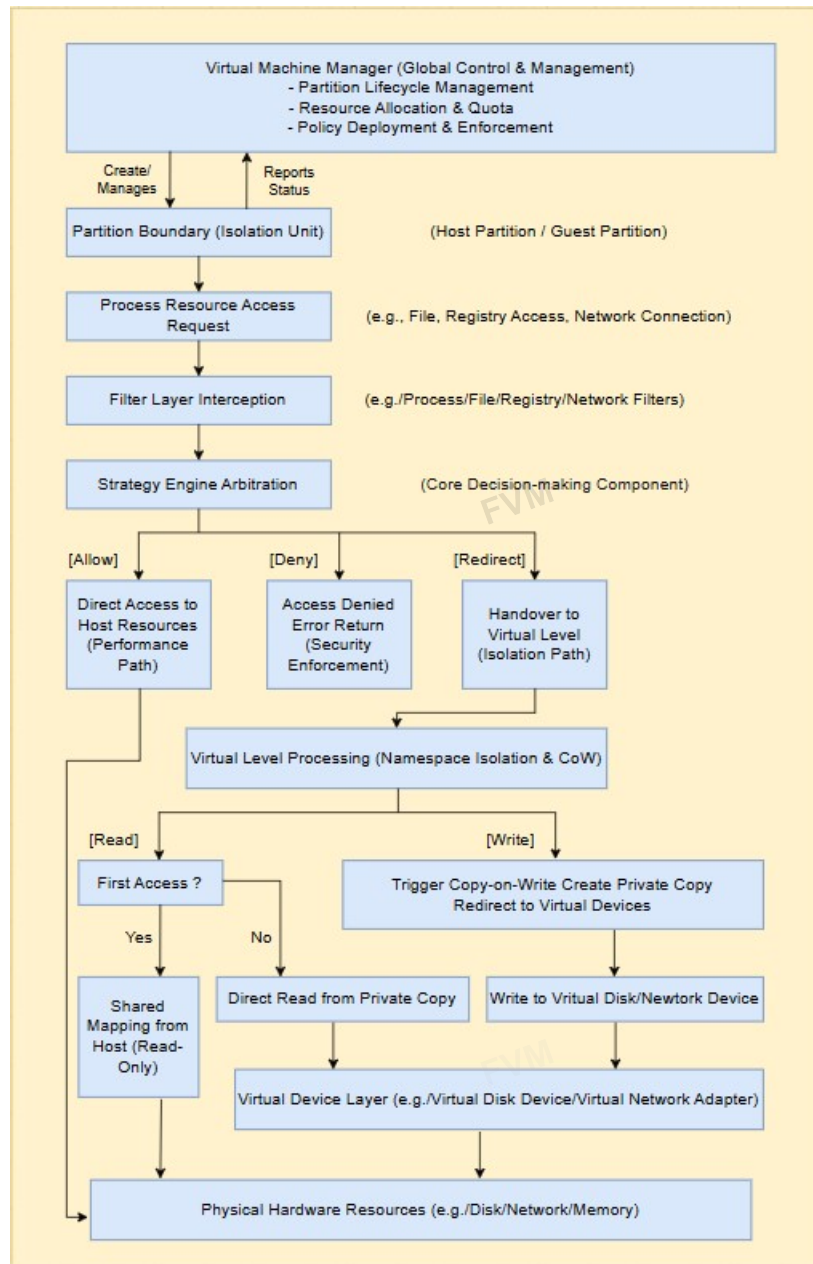
The FVM Virtual Device Layer consists of virtualized resource units built on top of physical hardware. By abstracting and emulating physical resources like disks and networks, it provides independent, isolated device views for each partition instance. Its core responsibilities include:

- **Virtual Disk Device:** The core implementation unit for FVM storage virtualization. Responsible for abstracting physical disk resources into multiple independent, isolated virtual storage views and providing dedicated disk access interfaces for each partition instance.
- **Virtual Network Device :** The core implementation unit for FVM network virtualization. Responsible for abstracting physical network resources into multiple independent, isolated virtual network views and providing dedicated network access interfaces and communication control capabilities for each partition instance.

The FVM Virtual Device Layer acts as the underlying execution unit for resource abstraction and isolation control, extending the security boundary of physical resources from the "kernel layer" to the "device layer." It provides trustworthy, controllable, and metered virtualized infrastructure for partition instances.

5. Logical Architecture

The diagram below provides a brief overview of the logical architecture of the FVM environment.



FVM's architecture is based on a layered decoupling and policy-driven paradigm, constructing a complete management and control loop from access triggering to resource execution. Its core logical relationship can be formalized as: **using partitions as isolation boundaries, the filter layer as sensing contacts, the strategy engine as the decision-making hub, the virtual layer as the execution agent, virtual devices as resource abstraction carriers, and the virtual machine manager as the global control entity.** The collaborative relationships and data flow paths among components are as follows:

5.1 Partition Assignment and Process Binding Mechanism

- **Partition Instantiation :** The Virtual Machine Manager, as the global control entity, creates and maintains the logical boundaries of partition instances, establishing the basic units for resource isolation.
- **Dynamic Affiliation Establishment:** The Process Filter monitors process creation events in real-time. By querying the Strategy Engine for affiliation based on predefined rules, it dynamically binds the process to a designated partition (host or guest). Once established, this affiliation becomes the baseline context for all subsequent access control decisions, persisting throughout the process's lifecycle.
- **Lifecycle Maintenance :** The Virtual Machine Manager continuously tracks partition states, ensuring the entire process from startup to termination is controllable and traceable, and maintaining the continuity of process affiliations during migration.

5.2 Intercept-Decide-Execute Semantics for Access Requests

When a process within a partition initiates a system call for resource access, the following standardized processing pipeline is triggered:

- **Interception Layer:** The corresponding functional filter (File, Network, Registry, etc.) intercepts the request before it reaches the original system resource stack, pauses the execution flow, and reports the request context to the Strategy Engine.
- **Arbitration Layer:** The Strategy Engine performs a multi-dimensional evaluation based on the partition identifier, resource type, security policies, and real-time status, outputting a clear disposition semantic: Allow (pass through to host), Deny (terminate access and return an error), or Redirect (hand over to the Virtual Layer for processing).
- **Execution Layer :** Executes the corresponding action based on the Strategy Engine's decision. For the "Redirect" semantic, the Virtual Layer activates namespace isolation and copy-on-write mechanisms to reconstruct the resource access path.

5.3 Virtualized Mapping of Resource Views

- **Read Operation Semantics :** The Virtual Layer, through a shared mapping mechanism, exposes public resources from the host partition (e.g., read-only system files) to guest partitions in read-only mode. When a guest partition accesses a shared resource for the first time, the Virtual Layer establishes a mapping from the host's physical page to the guest's virtual address space, achieving zero-copy resource sharing.

- **Write Operation Semantics :** Write operations trigger the copy-on-write mechanism. The Virtual Layer intercepts the write request, instantly creates a private copy of the target resource within the partition's dedicated storage domain (provided by the Virtual Disk Device), and redirects subsequent write operations to this copy. This mechanism ensures the immutability of the host's original resources while enabling on-demand storage allocation.
- **Metadata Operation Semantics:** For non-file resources like kernel objects, the Virtual Layer uses redirection technology to maintain the partition's independent virtualized view. All modification operations are isolated within the partition boundary.

5.4 Resource Abstraction and Proxy Relationship of the Virtual Device Layer

- **Logical Abstraction of Physical Resources :** Virtual Disk Devices and Virtual Network Devices act as logical proxies for underlying hardware resources. They abstract physical storage media and network interfaces into multiple independent, isolated virtual device instances for exclusive use by partitions.
- **Transparent Data Path Transformation:** While abstracting resources, the Virtual Device Layer also implements transparent transformation of the data plane, supporting advanced control capabilities like traffic mirroring, access blocking, and encrypted transmission, extending security boundaries from the kernel layer to the device layer.
- **Resource Metering Interface :** The Virtual Device Layer reports resource consumption data to the Virtual Machine Manager, providing underlying support for quota validation, billing, and auditing.

5.5 Closed-Loop Collaborative Relationship of the Management Plane

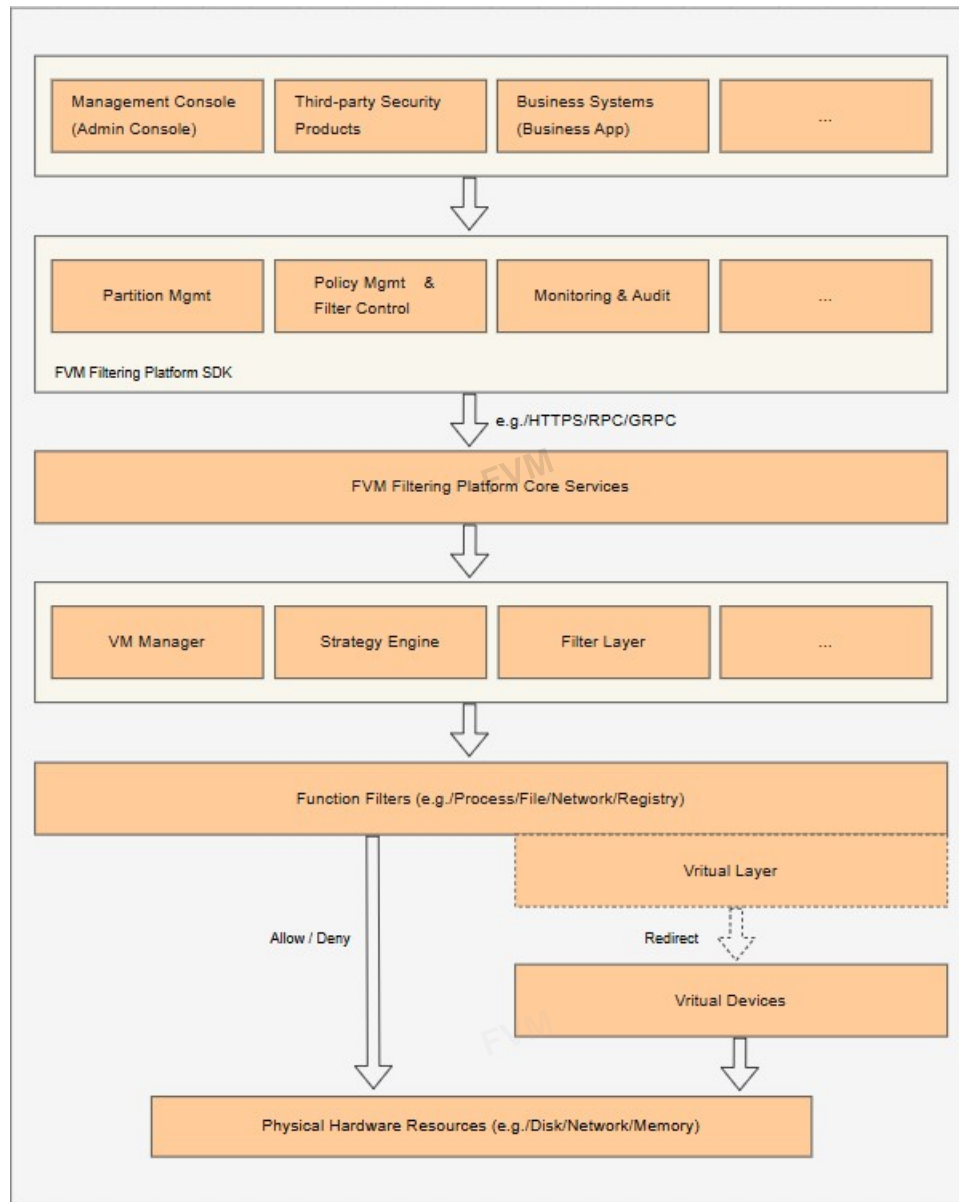
- **Complementarity of Static Quotas and Dynamic Policies:** The Virtual Machine Manager handles static resource allocation for partitions (e.g., CPU quota, storage limits), while the Strategy Engine manages the dynamic control of runtime behavior. This forms a dual-control mechanism of "resource definition" and "behavior constraint."
- **Consistency of Policy Deployment and Enforcement :** The Virtual Machine Manager uniformly deploys isolation policies to the Strategy Engine, which performs real-time arbitration based on these rules during runtime, ensuring that control intentions are consistently implemented across all system layers.
- **Full-Dimension Coverage and Audit Consistency :** Components like the MISC Filter complete the final management dimension, ensuring that control semantics for resources like desktop objects and system services are entirely consistent with those for files, networks, etc. This builds a comprehensive logical closed loop covering processes, storage, networks, and other resources within the FVM Filtering Platform.

- **Observability and Feedback Mechanism :** Each execution layer (Filter Layer, Virtual Layer, Virtual Devices) reports operational status and resource consumption data to the Virtual Machine Manager, forming a complete observability closed loop from control command issuance to execution effect feedback, supporting dynamic optimization and anomaly diagnosis.

The logical architecture of FVM is essentially an affiliation-determination-based full-path control model, orchestrated by the Virtual Machine Manager as the global control entity. From the moment a process is created, it is marked with a partition identity. Every subsequent resource access follows the standard semantic pipeline: Filter Layer sensing -> Strategy Layer decision -> Virtual Layer redirection -> Device Layer execution. Ultimately, with the guarantee of copy-on-write and namespace isolation mechanisms, efficient resource reuse and strong runtime isolation constraints are achieved. The Virtual Machine Manager's orchestration, deployment, monitoring, and adjustment functions throughout ensure the entire system's controllability, observability, and consistency.

6. Overview of the FVM Filtering Platform SDK

The diagram below provides a brief overview of the interaction flow between a caller and the underlying components via the FVM Filtering Platform SDK.



The FVM Filtering Platform exposes a set of layered, fully functional application programming interfaces (APIs) for upper-layer management tools, third-party security products, and business systems to call, enabling fine-grained control and deep integration with the virtualization environment. The SDK design follows the principles of **separation of duties** and **extensibility**, encapsulating underlying virtualization capabilities into standardized service interfaces.

6.1 Core Service Layer SDK

The Core Service Layer SDK provides direct management interfaces to the foundational capabilities of the FVM Filtering Platform, serving as the building block for upper-layer functions.

- **Process Management Interface :** Manages process-level isolation behaviors, configures process monitoring/interception/allowing rules, queries process partition affiliation, and migrates processes to designated partitions.
- **File Management Interface :** Manages file system access and virtualization, configures file monitoring/interception/allowing access rules, and sets virtual file views.
- **Registry Management Interface :** Manages registry access and virtualization, configures registry monitoring/interception/allowing access rules, and sets virtual registry views.
- **Network Management Interface:** Manages network access and traffic processing, configures network monitoring/interception/allowing access rules, configures DNS access rules, and controls transparent proxy behavior.
- **Kernel Object Management Interface :** Manages kernel resource access and virtualization, configures kernel object interception/allowing access rules, and sets virtual kernel object views.
- **Miscellaneous Management Interface :** Manages access to resources like services and desktops, configures service virtualization rules, and sets desktop object isolation policies.

6.2 Management and Orchestration Layer SDK

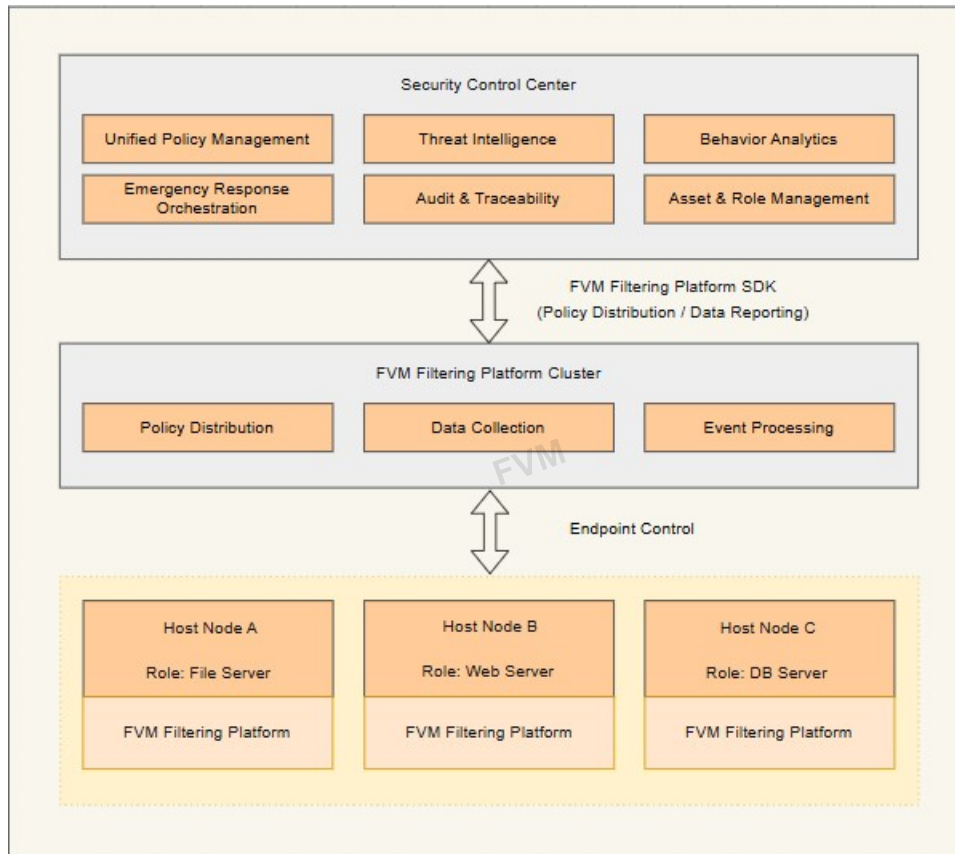
The Management and Orchestration Layer SDK provides unified control capabilities for the virtualization environment, targeting upper-layer orchestration systems.

- **Partition Lifecycle Management Interface :** Manages partition instances, creates/starts/stops/destroys partitions, and queries partition status.
- **Monitoring and Audit Data Interface:** Obtains system resource access runtime data, queries resource usage statistics, exports audit logs, retrieves real-time performance metrics, and monitors process resource access behavior.

The FVM Filtering Platform SDK, through its **layered design, functional classification, and standardized interfaces**, encapsulates system resource access control and virtualization capabilities into services that are easy to integrate and invoke, evolving from a closed resource access control engine to **an open programmable isolation platform**.

7. Host Security Protection System Based on the FVM Filtering Platform

The diagram below provides a brief overview of the host security protection structure of the FVM Filtering Platform.



The core idea of the FVM Filtering Platform is to build a unified resource access control framework covering all partitions. Using the FVM Filtering Platform SDK, the protection boundary can be extended from guest partitions to the host partition, achieving defense-in-depth for the physical host. This chapter elaborates on how to leverage platform capabilities to build a host security protection system, forming an overall protection network covering a cluster of hosts through role definition, policy customization, behavior monitoring, and threat interception.

7.1 Protection System Design Philosophy

Traditional host security protection relies on endpoint security software, which has limitations such as lagging protection, rigid policies, and isolated operation. The FVM Filtering Platform redefines the paradigm of host security protection through control capabilities built into the operating system kernel layer:

- **Secure by Default:** All resource accesses are subject to control by the filter layer; no default trust exists. Every file, registry, network, and other resource access

must undergo strict arbitration by the Strategy Engine, moving the security perimeter forward to every system call.

- **Role Differentiation :** Each host assumes a specific functional role and is configured with dedicated security policies. File servers focus on file access control, web servers focus on process integrity protection, database servers focus on data file tamper resistance, and domain controllers focus on protecting critical system resources. This achieves deep integration of security policies with business characteristics.
- **Centralized Management and Control :** Dispersed hosts are connected into a unified management plane via the SDK. Policies are defined centrally and deployed in batches, behavioral data is aggregated for analysis, and threat intelligence is shared in real-time, forming cluster-level collaborative protection capabilities.
- **Dynamic Response :** Based on continuous behavior monitoring, real-time discovery and rapid interception of unknown threats are achieved. Upon detecting anomalous behavior, targeted policies can be quickly formulated and deployed within seconds, blocking attacks in their early stages.

7.2 Host Role Definition and Policy Customization

In a cluster environment, each host assumes different functional roles. The FVM Filtering Platform allows defining a dedicated role for each host and customizing fine-grained resource access policies based on that role.

- **File Server :** Undertakes file sharing functions, with core business centered around file reading and writing. Faces threats like ransomware encryption and unauthorized access. Security policies focus on file access control and process whitelisting, ensuring only trusted processes can run, only trusted processes can write to shared directories, only ports 139 and 445 are open, and access source IPs are controlled.
- **Internet-Isolated Terminal:** These terminals are primarily for resource-intensive software (e.g., integrated development environments, compilers, large design software). Users need to develop, test, research, and design on the terminal, but direct exposure to the internet introduces risks like malware introduction and lateral movement. The core of the security policy is combining network isolation with access proxy: The terminal is prohibited from directly accessing the internet by default. All external network requests are processed by connecting to the FVM Cloud Desktop Virtual Application via an internet client. External traffic is fetched in the isolated environment, and only necessary content is returned, preventing external threats from directly touching the local terminal environment. For internal resource access, a network whitelist is used, allowing connections only to necessary resources like designated code repositories, dependency mirrors, and test environment servers, significantly reducing the network attack surface. Peripheral access permissions are

controlled to prevent malware introduction via physical media or sensitive code from being taken out of the environment.

- **Business Migration Terminal:** These terminals are primarily for handling internal business tasks like email, document processing, outsourced business system access, and software development. Users frequently handle sensitive business data, facing threats like phishing emails, malware infections, and data leaks. The core security policy is isolating business data by running enterprise business applications in independent guest partitions via the FVM platform, completely isolating business data from the personal environment. The entire business application runtime environment is isolated from the host machine via the application partition. All business data is stored within the partition, invisible to the host's personal files, browsing history, and temporary files. If the terminal gets infected with malware, the attacker is confined to the host environment and cannot access business data within the partition. When a device is decommissioned or an employee leaves, simply destroying the guest partition ensures no enterprise data remains, preventing business data leakage.

7.3 Threat Discovery Based on Behavior Monitoring

- **Full-Dimension Data Collection:** Continuously collects behavioral data such as process creation and termination, file operations, registry operations, network connection establishment, kernel object access, and peripheral connections, reconstructing the complete execution chain.
- **Anomaly Behavior Detection Models:** Build models like bulk file modification (for ransomware encryption), anomalous process chains (for phishing attack characteristics), rare domain name resolution (for C2 communication), sensitive file reading (for data exfiltration attempts), anomalous port connections (for reverse shells), registry persistence (for backdoor installation), peripheral connection (for unauthorized storage device usage), cross-partition access (for boundary breach attempts), network whitelist deviation (for unauthorized internal resource connections), and traffic proxy deviation (for processes bypassing proxy to connect directly to the internet).
- **Monitoring and Alerting Closed Loop:** Behavioral data is reported in real-time to the security management center. The analysis engine executes rule matching and baseline comparison in parallel, triggering alerts when threats are hit or deviations are found, which are pushed to security analysts for investigation.

7.4 Threat Interception Based on Policy Deployment

- **Rapid Response Process:** After threat investigation, policies based on the three core semantics (allow, deny, redirect) are formulated. After simulation and validation, they are deployed in batches to affected hosts within seconds to block the threat in real-time. Effectiveness is evaluated, and optimizations are made.

- **Typical Interception Scenarios:** Block unknown processes from writing to shared directories during a ransomware outbreak; block processes from connecting to mining pools during cryptominer propagation; block unauthorized processes from reading sensitive files during data exfiltration attempts; block connection requests from anomalous IPs during lateral movement attacks; immediately block device mounting and trigger alerts upon unauthorized peripheral connection; intercept and alert on processes configured for proxy access attempting direct internet connections on internet-isolated terminals; when a business migration terminal is infected with malware, the attacker cannot reach business data as business apps run in isolated partitions.

7.5 Forming an Overall Protection Network

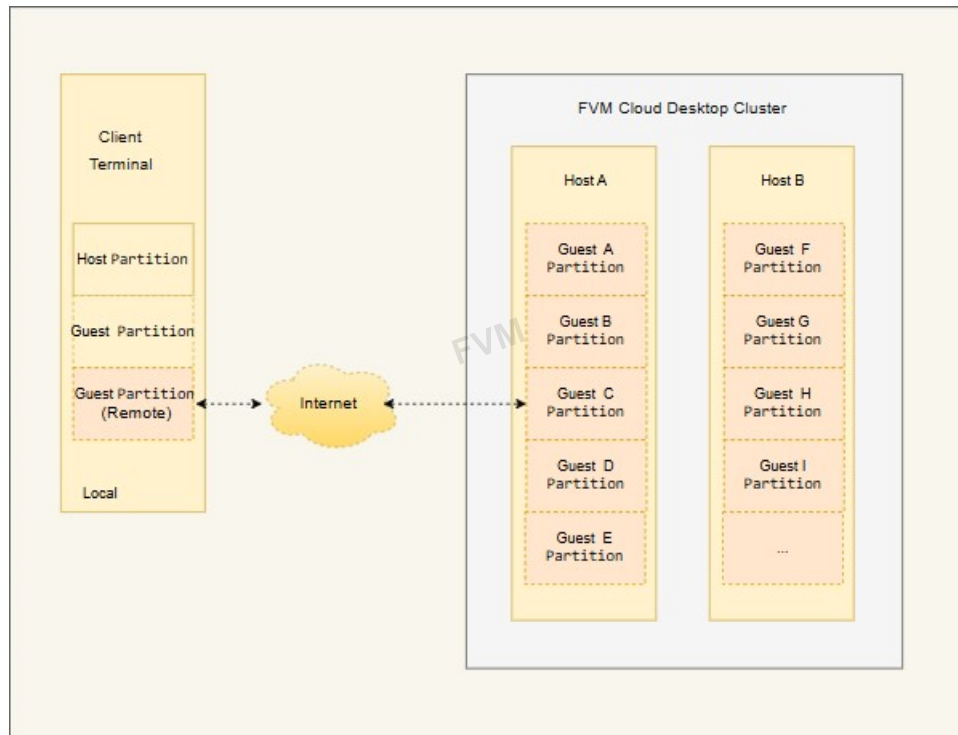
- **Three-Layer Protection Architecture :** The upper layer is the Security Management Center (decision hub), the middle layer is the Management Cluster (neural transmission system), and the lower layer comprises Host Nodes (execution units). Each host runs the complete FVM Filtering Platform, including various filters for processes, files, networks, and the registry.
- **Collaborative Protection Capabilities:** Centralized policy deployment takes effect in seconds; real-time threat intelligence sharing prevents lateral movement; aggregated behavior analysis uncovers hidden attacks; cross-host correlation reconstructs the full attack chain; automated response orchestration reduces incident response time; unified peripheral control policies prevent malware propagation across hosts via physical media; centralized network whitelist management ensures compliant access to internal resources; centralized partition policy management ensures standardized implementation of business data isolation.
- **Collaborative Workflow :** Host A detects an anomaly and reports it to the management center. Analysis confirms a virus. The management center generates an interception policy and deploys it in batches to all hosts. The entire network simultaneously gains the ability to intercept, blocking virus propagation.

The host security protection system based on the FVM Filtering Platform, guided by the four core principles of secure by default, role differentiation, centralized management, and dynamic response, evolves host security from passive defense to active protection, building a defense-in-depth system that covers the entire enterprise cluster. Within this framework, the process whitelisting and isolated execution mechanisms on service nodes, the classified protection (internet-isolated and business migration types) on terminal nodes, and the fine-grained control of peripheral management mechanisms work in synergy. This not only curbs the possibility of malicious code execution at the source but also establishes a security protection model deeply tailored to specific usage scenarios for terminal nodes. Together, they form a complete, multi-layered active

defense system that ensures the safe and stable operation of enterprise businesses in an increasingly complex threat environment.

8. FVM Cloud Desktop Virtual Application

The diagram below provides a brief overview of the interaction between a client and the cloud desktop virtual application.



The FVM Cloud Desktop Virtual Application is a multi-user desktop sharing solution built on FVM lightweight virtualization technology. This solution creates multiple mutually isolated virtual desktop environments on a single physical host, delivering a complete desktop experience to end-users via remote connection protocols. Unlike traditional Virtual Desktop Infrastructure (VDI), FVM Cloud Desktop uses process-level virtualization and copy-on-write technology, eliminating the need to load a complete operating system image for each user. It achieves second-level deployment of desktop environments, efficient resource reuse, and centralized data management. All user applications run within FVM guest partitions on the host machine, while endpoint devices merely serve as interactive interfaces for display output and input capture. Business data resides entirely on the cloud server, fundamentally eliminating the risk of data leakage on the endpoint side.

- **Data Resides in the Cloud:** All business data, user profiles, and application data are stored on cloud servers. Endpoint devices do not retain any copies of business data. Even if an endpoint device is lost, stolen, or infected with

malware, enterprise sensitive data remains securely stored within the data center.

- **Efficient Resource Reuse :** Based on copy-on-write technology, multiple users share the same read-only operating system and application template, with only personalized delta data recorded per user. Memory and CPU utilization are significantly improved.
- **Balance Between Isolation and Sharing:** Each user runs within an independent FVM guest partition. Namespace isolation achieves complete separation of processes, files, registry, and networks, while copy-on-write allows sharing of read-only resources, maximizing resource utilization while ensuring security.
- **Access Proxy Capability :** The Cloud Desktop Virtual Application can act as a secure proxy for endpoints accessing the internet. Endpoints are prohibited from directly accessing external networks by default; all internet traffic is forwarded through the cloud desktop guest partition. Content is fetched and threats are filtered in the cloud, with only necessary screens returned, preventing external threats from directly touching the local endpoint environment.

9. FVM Enterprise and Personal Data Protection Capabilities

The diagram below provides a brief overview of the FVM data protection capability landscape.



The FVM data protection system, based on three core mechanisms—partition isolation, resource control, and data traceability—builds an all-around data security defense for enterprises and individual users. Enterprise data is only usable within the FVM-controlled environment; it becomes inaccessible outside this environment. When data needs to be shared externally, it can be exported in PDF or FVM-specific formats, catering to read-only distribution and secure collaboration needs respectively. Dual-layer protection with visible and invisible watermarks ensures traceability throughout the entire lifecycle.

9.1 Core Data Protection Mechanisms

- **Business Data Isolation in Guest Partitions:** All enterprise business applications run within independent FVM guest partitions. Business data exists only inside these partitions, completely isolated from the host's personal environment. If an endpoint gets infected with malware, the attacker is confined to the host environment and cannot access business data within guest partitions. When a device is decommissioned or an employee leaves, simply destroying the guest partition ensures no enterprise data remains, fundamentally solving the endpoint data leakage problem.
- **Security Control in Host Partition :** The host machine, as the base runtime environment, has strict resource access controls enforced by the FVM Filtering Platform. This ensures only trusted processes can run; protects critical system files and registry keys from tampering; and restricts the host partition's network access permissions to prevent it from becoming an attack pivot. The security control of the host partition provides a trusted runtime foundation for upper-layer guest partitions.
- **No Data Residency in Cloud Desktop Virtual Application :** Through the FVM Cloud Desktop Virtual Application, users can remotely access enterprise business systems from endpoints. All business data resides entirely in the cloud, with endpoint devices only receiving encrypted screen image data. Even if an endpoint device is lost or stolen, enterprise sensitive data remains securely stored within the data center, achieving true data non-residency.
- **Controlled Export and Secure Transfer:** Enterprise data is only usable within the FVM-controlled environment. Any attempt to take data out of this controlled environment must go through the controlled export mechanism. Supported export formats: PDF format for external read-only distribution, automatically adding visible and invisible watermarks; Image/Video format for sharing materials, automatically adding visible and invisible watermarks; FVM-specific format for internal secure collaboration, featuring encrypted file protection and embedded invisible watermarks, accessible only by authorized partitions.

9.2 Business Data Isolation in Guest Partitions

- **Business Applications Running in Partitions:** Enterprise business applications are migrated via the FVM platform to run in independent guest partitions. Business data exists only within the partition, achieving complete isolation between business data and the personal environment.
- **Partition Data Lifecycle Management:** Business data within a guest partition is tied to the partition's lifecycle. Upon user login, the partition uses a user-specific encrypted disk. Upon user logout, delta data is saved to the encrypted disk, and partition resources are reclaimed. For temporary tasks or external personnel access, partitions can be configured to auto-destroy along with all data after the session ends, ensuring no data residue.

- **Cross-Partition Access Control :** FVM prohibits direct data access between partitions by default, ensuring data independence for each partition. For scenarios requiring data exchange across partitions, controlled transfer is achieved through the encrypted FVM file format, preventing unauthorized data access.

9.3 Partition Security Control

- **Trusted Process Whitelist:** Enforces strict process whitelisting within partitions via the FVM Filtering Platform, allowing only approved processes to run. Any attempt to start an unauthorized process is intercepted, fundamentally blocking the possibility of malicious code execution in the controlled environment.
- **Protection of Critical System Resources:** Protects critical resources in both host and guest partitions via the FVM Filtering Platform. Access to key locations like system directories, registry configuration items, and startup items is restricted to trusted processes only. Attempts by unauthorized processes to modify system files or write to auto-start registry keys are immediately intercepted, and alerts are triggered.
- **Screen Watermarking and Screenshot Prevention:** Provides configurable screen watermarking and screenshot prevention capabilities for all partitions via the FVM Filtering Platform.
- **Network Access Restriction :** Implements fine-grained control over network access permissions for all partitions via the FVM Filtering Platform. Network policies can be configured independently for each partition: Host partitions might only access necessary management channels, update servers, and security management centers by default; Guest partitions can be configured to access only specific internal servers or the internet via a proxy based on business needs. Network policies are bound to the partition identity, ensuring consistent access permissions even after partition migration or recreation.

9.4 No Data Residency in Cloud Desktop Virtual Application

- **Zero Data Caching on Endpoint:** During a cloud desktop session, the endpoint device does not store any copies of business data. Even if temporary files or cached data are generated during the session, they are entirely stored within the cloud session. No business data residues exist on the endpoint's local disk or memory.
- **Cloud Desktop as a Secure Internet Proxy:** The cloud desktop itself runs within an FVM guest partition and is uniformly managed by the FVM Filtering Platform, inherently possessing the same level of security protection as local guest partitions. When an endpoint needs to access the internet, the cloud desktop can serve as a secure proxy, building a complete end-to-end protection system. Local endpoint guest partitions are prohibited from directly accessing the internet by default; all internet requests are forwarded to the cloud desktop.

Browsers or applications within the cloud desktop initiate the actual internet access requests. The traffic is inspected by the network filter in the cloud desktop's partition and is only forwarded through the internet egress after conforming to the policies.

The local endpoint guest partition is protected by the FVM Filtering Platform. The remote connection channel between the endpoint and the cloud desktop is encrypted. The cloud desktop partition is again protected by the FVM Filtering Platform and performs deep inspection on internet traffic. External threats cannot directly reach the local endpoint environment and struggle to penetrate the cloud desktop's multiple layers of protection. Simultaneously, even if the local endpoint is infected with malware, because internet access must go through the cloud desktop proxy, the malware cannot bypass controls to transmit data outward directly.

9.5 Controlled Export and Secure Transfer

Enterprise data is only usable within the FVM-controlled environment. Any attempt to take data out of this controlled environment must go through the controlled export mechanism.

- **PDF Format Export - External Read-Only Distribution:** When sharing data with external partners, clients, or regulatory bodies, documents can be exported as PDF. Exported PDF files are forced into read-only mode. Recipients can only view the content and cannot modify or redistribute it. Visible watermarks are automatically overlaid, and invisible watermark traceability information is embedded.
- **Image/Video Format Export - External Distribution :** When sharing data with external partners, clients, or regulatory bodies, export as images or videos is supported. Exported files automatically have visible watermarks overlaid and invisible watermark traceability information embedded.
- **FVM File Format Export - Internal Secure Collaboration:** When transferring data between different departments, hosts, or partitions within an enterprise, exporting to the FVM-specific encrypted file format is supported. This ensures secure and efficient data transfer while ensuring security.

9.6 Enterprise Data Protection Scenarios

- **Endpoint Data Protection :** On an endpoint, business systems are accessed via guest partitions, isolating business data from the personal environment. When needing to share a work document, export it as PDF; the recipient can only view, not modify or copy. When collaborating internally on a file, export it as an FVM file, accessible only by authorized partitions. When needing to access the internet for research, use the cloud desktop proxy; all traffic goes through cloud

security checks, and the local endpoint doesn't need a direct internet connection, reducing the attack surface.

- **Source Code Protection:** Development source code is stored in guest partitions. Design documents for external review are exported as PDFs. For cross-department collaboration, source code is exported as FVM files.
- **Compliance Audit Scenario:** An enterprise needs to submit an audit report to a regulatory body. The report is exported as a PDF, with files automatically containing visible and invisible watermarks of the recipient's information and set to read-only protection. Even if the file is secondarily disseminated after the regulatory body reviews it, the invisible watermark allows tracing back to the initial recipient, clarifying leak liability.

9.7 Personal Data Protection Scenarios

- **Personal Privacy Isolation:** Individual users can create multiple guest partitions on the same device for different scenarios like work, personal life, and finance. Data in each partition is completely isolated. Documents and emails in the work partition are invisible to photos and chat logs in the personal life partition, preventing cross-leakage.
- **Encrypted Storage of Sensitive Information :** Individuals can store highly sensitive information like bank statements, ID photos, and password files within encrypted guest partitions. Even if the device is stolen, attackers cannot decrypt and access the data within the partition.
- **Secure File Sharing:** When an individual needs to send a sensitive file to friends or family, they can choose to export it as a PDF, adding visible/invisible watermarks with recipient information, or export it as an FVM file, setting a password or time window for decryption. The watermark embedded in the file can record recipient information, allowing traceability if the file is secondarily disseminated.

9.8 Data Protection Audit and Traceability

- **Full-Link Audit Logs:** The FVM Filtering Platform generates detailed audit logs for all data access behaviors across partitions, including file access logs, partition creation/destruction logs, export operation logs, decryption attempt logs, dark watermark embedding records, and network connection logs. Logs contain complete context including timestamps, host information, user identity, partition identifier, process information, operation type, and operation object.
- **Anomaly Behavior Detection :** Based on audit logs, the security management center can build anomaly behavior detection models to identify potential data leak risks. Examples include detecting a user exporting a large number of PDF or FVM files in a short time; detecting a user frequently attempting to decrypt unauthorized FVM files; detecting exported files being sent to anomalous external email addresses; detecting a partition suddenly initiating a large

number of network connections to unknown targets. Alerts are automatically triggered upon detection, and interception can be coordinated.

- **Data Leak Traceability:** When a data leak incident occurs, audit logs and dark watermark technology work together to achieve precise traceability. After obtaining the leaked file, the security team extracts the dark watermark to get information like export time, operator, and recipient. Combined with audit logs, they can reconstruct the complete path from export to leak, identifying the leak point and responsible party, providing evidence for incident handling and legal accountability.

FVM builds an all-around data protection system for enterprises, covering endpoint to cloud, storage to transmission, and internal to external. It also provides privacy isolation and sensitive information protection capabilities for individual users. Enterprise data is only usable within the FVM-controlled environment; it becomes inaccessible outside. When transferred externally, controlled export and dark watermark technology enable precise traceability, ensuring data security remains controllable while facilitating efficient data flow.

10. FVM Software Multi-Instance and Virus Isolation Capabilities

The diagram below provides a brief overview of the FVM software multi-instance and virus isolation capability landscape.



Based on partition isolation and resource control mechanisms, the FVM Filtering Platform provides a lightweight, highly secure solution for running

multiple instances of software and isolating viruses on the same physical host. Whether needing to run multiple instances of the same application simultaneously or wishing to isolate unknown programs from the system environment, FVM achieves process-level isolation through independent guest partitions, ensuring applications do not interfere with each other and viruses cannot spread.

10.1 Software Multi-Instance Capability

Traditional software multi-instance operation faces numerous challenges: Many applications restrict a single user to running only one instance through mechanisms such as mutexes and registry keys; when multiple instances run simultaneously, configuration files and cached data may overwrite each other, leading to conflicts. FVM fundamentally solves these problems by creating independent runtime environments for each application instance through partition isolation mechanisms.

- **Multi-Instance Application Scenarios:** In gaming multi-instance scenarios, when users need to run multiple game accounts simultaneously, an independent guest partition can be created for each account. In IM multi-instance scenarios, independent partitions are created for each account, allowing simultaneous login to WeChat, QQ, etc. Processes within each partition run independently, with chat histories and cached files stored in isolation, ensuring work and personal data do not interfere with each other, meeting multi-identity management needs. In development and testing multi-environment scenarios, when developers need to simultaneously test the behavior of the same application under different configurations, independent partitions are created for each test environment. Each partition has its own configuration file view and environment variables, ensuring test results do not affect each other, enabling parallel validation of multiple configuration combinations.
- **Resource Optimization and Sharing:** Although partitions run independently, efficient resource sharing is achieved through copy-on-write technology. Read-only resources such as core operating system files, public DLLs, and application executable files are shared in physical memory across all partitions, adding almost no additional overhead. Only incremental content such as configuration files and cached data for each partition is stored independently, occupying disk space on demand. This design makes it possible to run dozens of application instances on a single physical host, with resource consumption far lower than traditional virtual machine solutions.

10.2 Virus Isolation Capability

In traditional security models, users face the risk of system infection when running unknown programs. Through partition isolation mechanisms, FVM can isolate unknown programs, suspicious files, and high-risk operations to run in

independent environments. Even if a program contains malicious code, it cannot breach the partition boundary to threaten the host system.

- **Sandbox Partition Isolation :** When needing to run programs from unknown sources, open suspicious attachments, or visit high-risk websites, they can be isolated and executed in a sandbox partition. The sandbox partition is completely isolated from the host partition and business partitions, with malicious behavior confined within the partition. Sandbox partitions are temporary in nature—they can be created on demand and automatically destroyed after the session ends, preventing malware from persisting in the system. Sandbox partitions share the host's read-only system files, with all write operations redirected to temporary storage through copy-on-write mechanisms, ensuring the host system remains unaffected by any modifications. At the network level, sandbox partition access permissions can be restricted, blocking communication between malicious programs and command-and-control servers; at the peripheral level, access to physical devices such as peripherals is prohibited by default, preventing malware propagation through peripherals.
- **Malware Behavior Interception :** Even when malicious programs run within a sandbox partition, the FVM Filtering Platform continuously monitors and intercepts their behavior. When ransomware attempts to bulk-modify files, the file filter can intercept its encryption activities—files within the sandbox partition may be encrypted, but external real data remains unaffected. When cryptomining programs attempt to connect to mining pools, the network filter can intercept their network communications—the mining process may run but cannot generate actual revenue. When backdoor programs attempt to write to auto-start registry keys, the registry filter intercepts their write operations—the backdoor disappears when the sandbox is destroyed. When malware attempts to scan the internal network for lateral movement, the network filter can restrict its intranet access permissions, blocking the propagation path.
- **Malware Analysis Environment:** Security analysts can run suspected malicious samples in isolated partitions, observing their behavior without compromising the analysis environment. The FVM Filtering Platform comprehensively records all behaviors of the sample, including file operations, registry operations, network connections, process creation, and memory access, generating complete behavior analysis reports. Analysts can determine the nature of the sample based on these reports and formulate corresponding defense strategies. This analysis environment can be reused repeatedly—destroying the partition after each analysis restores a clean state, eliminating the need to reinstall the system or virtual machines.
- **Browser Isolation :** For the high-risk scenario of web browsing, FVM provides browser isolation capabilities. Users run browsers through isolated partitions, with all web content rendered within the partition. Even when visiting malicious websites containing exploit code, attacks cannot breach the partition boundary. In standard isolation mode, the browser runs in an independent partition, and

web pages cannot access the host file system; in deep isolation mode, functions such as downloads, clipboard access, and plugins can be further disabled, preventing malware downloads through the browser; in one-time isolation mode, the partition is automatically destroyed when the session ends, ensuring malicious cookies and cached data do not persist; in network isolation mode, the browser can be restricted to accessing only specific sites, preventing access to phishing websites.

- **Document Isolation:** For email attachments and documents received via instant messaging, FVM supports isolated document opening. When users open suspicious documents, the document viewer or editor launches within an isolated partition. Malicious macros and exploit code within the document are confined to the partition and cannot infect the host environment. Even if a document carries ransomware, its encryption activities only affect temporary files within the partition—all effects are eliminated when the partition is closed.

The three core mechanisms—partition isolation, efficient resource sharing, and comprehensive behavior monitoring—work in synergy, enabling FVM to achieve secure multi-instance operation and isolated execution on the same physical host. This not only meets user demands for parallel multi-instance operation but also ensures absolute system security when facing unknown threats.